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# Semantic BIM enrichment using a hybrid ML and rule-based framework for automated tenement compliance checking

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ABSTRACT

Semantic enrichment enhances BIM models by extracting structured information, improving their applicability for Automated Code Compliance Checking. Rule-based methods rely on well-defined conditions but struggle with tasks like space classification, where explicit checking rules are unavailable. Meanwhile, ML-based classification introduces adaptability but faces liability challenges due to misclassifications. This paper proposes a hybrid framework integrating ML for space classification and rule-based inferencing for tenement identification. The approach ensures that ML automates preprocessing, improving classification through meticulous feature engineering, while rule-based reasoning guarantees logical consistency during verification. Validated using real-world datasets from residential projects in Mumbai, India as a case, the ML-based space classification component achieves an F1-score of 0.85 and accuracy of 0.86, demonstrating its effectiveness. The deterministic tenement identification process delivers error-free results for various dwelling configurations, making it highly suitable for verification workflows. This study advances scalable BIM-based compliance systems by refining semantic enrichment methodologies for future applications.

## 1. Introduction

The evolution of the construction industry is increasingly driven by advancements in digital technologies, with Building Information Modeling (BIM) at the forefront of this transformation. BIM has revolutionized the planning, design, and management of buildings by providing a digital representation of the physical and functional aspects of construction projects [1]. The promise of BIM extends beyond mere visualization, offering opportunities to automate processes, reduce errors, and ensure compliance with local building codes. However, the reality is that the industry's journey toward fully automated code compliance checking (ACCC) is still in its infancy [2]. Despite its critical importance, the code compliance process remains largely manual and labor-intensive, involving repeated back-and-forth interactions between architects, engineers, and regulatory bodies. The inefficiencies in this process hinder productivity and increase the risk of delays and errors in construction projects [3]. BIM offers the potential to automate compliance checking by incorporating building regulation requirements directly into BIM models. However, achieving this vision requires more than just model visualization; it necessitates a deeper semantic understanding and enrichment of BIM elements.

The requirement of semantic enrichment becomes particularly evident in the preconstruction phase, where critical design decisions regarding regulatory compliance need to be made. The user-defined data available at the preconstruction stage is generic and approximate. Hence, without enrichment, the raw data may lead to inefficiencies in the ACCC process. Architects, who are responsible for ensuring that their designs adhere to regulatory standards, often face challenges to explicitly provide the information required at the early stages of the project. This problem occurs due to the paradigm transfer of CAD to BIM mindset and the human errors in labor-intensive processes [4]. As a result, without enrichment and machine-readable representation of design code knowledge, the existing automatic code compliance checking systems are limited to handling a narrow set of code requirements, focusing primarily on explicitly defined model attributes or numerical constraints [5,6]. Therefore, there is a pressing need to enrich BIM models early in the design process, ideally during the initial modeling phases, to enable effective automated compliance checking.

The enrichment must be done in a manner that supports both rule-based logic and learning-based inference, leading to the need for a hybrid approach. A hybrid approach to semantic enrichment combines rule-based methods with machine learning techniques to leverage both

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strengths. Rule-based methods rely on predefined sets of conditions that must be met, which is particularly useful for handling building codes that are well-defined and deterministic [7]. However, not all aspects of building design can be effectively addressed through rule-based approaches. This is where machine learning comes into play, offering the ability to recognize patterns and make decisions based on data-driven insights [8]. Tenement identification, the process of categorizing and grouping spaces into functional residential units, exemplifies the need for such a hybrid approach. It involves both room classification, a prerequisite for understanding space types, and space association, which determines unit boundaries based on regulatory conditions.

Room classification plays a crucial role in tenement identification, as compliance standards vary across space types [9]. For instance, a bedroom must comply with specific natural light and ventilation standards, whereas a toilet must adhere to specific sizes for accessibility requirements. Therefore, identifying each space type is a primary step in enabling automated compliance checking. Room classification can be effectively approached as a machine learning problem, wherein algorithms are trained to recognize and classify spaces based on their features, such as dimensions and spatial relationships [10]. On the other hand, the premium cost of building and parking requirements for a tenement unit depends on the carpet area size. The space association tasks for identifying such tenements can be approached through logical reasoning [11]. This multi-stage enrichment process provides the foundation for more complex tasks, such as tenement identification and code compliance.

Adopting machine learning (ML) for semantic enrichment enhances the end user's ability to perform additional enrichment tasks quickly and in an automated manner. Further, ML addresses one of the key limitations of current ACCC systems — the inability to understand complex patterns and variations. Building codes often include provisions that require subjective interpretation, such as ensuring that spaces are 'adequately ventilated' or 'easily accessible'. While rule-based methods struggle with such ambiguous requirements, machine learning models can use historical data and examples to infer information required for compliance checking [12]. However, where rule-based systems do not classify objects slightly differing from the rulesets, ML-based systems classify all objects leading to false positive and false negative predictions [13]. False positive predictions are more detrimental than false negatives. For example, an ML model can lead to the approval of a serious design issue through false positive predictions in case of a compliance check. On the other hand, ML, as a user-aid, gets the predicted results verified for the rule-based checking at the reviews' end, removing the possibilities of erroneous verification liabilities from the urban local bodies (ULBs) [14]. This hybrid methodology enhances ACCC systems' reliability and makes them more user-friendly, accommodating, and organically adaptive to the industry.

This study focuses on semantic enrichment for tenement identification, addressing the challenge of grouping spaces into functional residential units based on spatial and regulatory constraints. Space classification is an inclusion task within tenement identification, as accurate labels are essential for tenement spaces before dwelling units are formed. The hybrid framework integrates ML for automated space classification and rule-based inferencing for tenement identification, improving accuracy and efficiency in compliance verification.

This research contributes to the field by introducing a hybrid approach for BIM semantic enrichment, developing a scalable feature extraction framework aligned with OpenBIM standards, validating the methodology using real-world residential datasets, and analyzing the role of feature selection in classification performance. The study aims to enhance automated tenement identification by refining space classification, implementing rule-based unit formation, optimizing feature selection, and advancing semantic enrichment for regulatory applications. The remainder of this paper presents a literature review on BIM semantic enrichment and hybrid classification methods, followed by a detailed methodology covering dataset preparation, feature engineering, and model implementation. Experimental results are then discussed, and model performance and tenement identification accuracy are evaluated. Finally, the paper concludes with a discussion of findings, limitations, and future research directions.

## 2. Background

This section reviews the current state of semantic enrichment in BIM, emphasizing its relevance to automated compliance workflows. It begins with a discussion of the limitations of existing BIM representations in supporting machine-readable reasoning, followed by an overview of object classification methods and their role in enriching model semantics. The section also highlights the evolution of rule-based and ML-based approaches. Finally, it outlines the existing research gaps that motivate the development of the proposed hybrid framework.

### 2.1. Semantic enrichment of building information models

The adoption of BIM in the AECO industry has revolutionized workflows, offering significant benefits such as enhanced collaboration and data integration. However, integrating BIM has also introduced challenges, particularly related to interoperability between stakeholders and the diverse platforms they employ [2]. The Industry Foundation Classes (IFC) schema was developed by BuildingSMART, offering a standardized format for data exchange to address this issue [15]. While IFC serves as a standard format, its complexity and limitations in semantics present obstacles for applications requiring advanced analysis, such as automated code compliance checking and building performance evaluations [16]. If regulation-specific object models were created depending upon the existing rules, more than three hundred and fifty semantic entities are required in IFC [17]. Therefore, a preprocessing layer must make a BIM/IFC model sufficiently enriched with data for machine readability. Even though IFC 4.1 contains 768 entities and 410 property sets, they are not trivial enough to address all required semantic classes [18].

Semantic enrichment involves augmenting BIM models with structured, meaning information so that building components can be interpreted beyond their geometric and spatial characteristics. Eventually, it facilitates the extraction of implicit information from the model, enabling automated reasoning, thus bridging the gap between design elements and building regulation requirements [13]. This process involves leveraging domain expertise and artificial intelligence to derive and embed implicit information into the model, thereby increasing interoperability and functionality for receiving applications [19]. For instance, researchers proposed the FORNAX platform, which encapsulates the semantics of building components, mainly focusing on attributes critical for code compliance. Their work demonstrated how semantic enrichment could bridge the gap between complex regulations and BIM models, enabling more effective automated reasoning and application [20].

The use of semantic web technologies has been another key development in this area. Unlike IFC, which primarily handles explicit information, semantic web approaches allow for the interrelation and reuse of diverse types of information [21]. This methodology was advocated, highlighting its ability to enable reasoning engines to derive implicit information through logical rules. Such innovations underscore semantic technologies' role in enhancing BIM models' adaptability to evolving industrial demands.

Research has further categorized semantic information into two broad types: individual element semantics and inter-element relationships. This distinction facilitates targeted enrichment strategies that address specific application needs [22]. For instance, semantic enrichment techniques have been extended to incorporate multidisciplinary methods, including machine learning and semantic web technologies, to address various challenges, such as damage detection in infrastructure and the development of City Information Models [23].

According to the broad implementation approach, semantic enrichment methods can be divided into semantic web technologies, rule-based reasoning, machine learning, and database-based semantic integration [24]. On the other hand, semantic enrichment can also be achieved by classifying the implementation into four different task types based on BIM object concepts and their property features [8]. Despite these advances, challenges remain in fully realizing the potential of semantic enrichment. The process of exporting models from native BIM tools to open formats such as IFC often leads to the loss of associational and topological relationships. This limitation hampers the ability of generic model review systems to process complex code requirements involving connections and aggregated data structures [14]. Consequently, researchers continue to explore more robust methods of integrating enriched semantics into BIM models to ensure consistency and utility across diverse platforms.

Semantic enrichment is pivotal for advancing BIM technologies and addressing critical gaps in interoperability and usability [25]. By enabling automated reasoning and bridging the semantic deficiencies of traditional formats, it aligns with the broader objectives of the AECO industry to achieve more intelligent, adaptable, and efficient workflows. Object Classification is a foundational semantic enrichment task, ensuring accurate representation and functionality of BIM elements. The classification of objects plays a crucial role in enabling enriched models to meet diverse application requirements.

### 2.2. BIM object classification

Object classification in BIM is essential for enhancing the semantic richness and utility of models. The loss or absence of semantic data during BIM model exchanges often necessitates reclassification, affecting subsequent applications such as design analysis and code compliance checking [13]. Misclassification or incomplete representation of objects can introduce inaccuracies in decision-making processes.

Early semantic enrichment approaches relied on expert systems that applied predefined rules to classify objects, particularly in domains such as bridge construction [26]. While achieving high accuracy in well-structured tasks, these rule-based systems lacked scalability and adaptability, as they required extensive rule sets tailored to specific object types [27]. Maintaining and updating such systems proved labor-intensive and impractical for broader applications.

Recent advancements have integrated ML and deep learning techniques to improve BIM object classification. The use of SVM and 3D deep learning networks demonstrated high accuracy in classifying objects based on IFC data [28,29]. However, computational demands and significant hardware requirements posed challenges for practical implementation [10]. Feature-based classification methods leveraging geometric and contextual attributes have also been explored. Combining geometric analysis with topological checks has improved the differentiation between constructive elements and clutter [30]. Additionally, logic-based inference rules using pairwise relationships have been effective in identifying objects in BIM models [31]. A combination of rule, ML, and Deep Learning-based ensemble approaches was proposed, highlighting the improvement in BIM MEP object classification capability [32].

A shift toward data-driven techniques has further advanced BIM object classification. The use of NLP for feature extraction from IFC files and the application of graph convolutional networks (GCNs) to classify BIM elements based on geometric properties has demonstrated potential [33,34]. However, most GCN applications fail to incorporate inter-object relationships, which are essential for semantic enrichment. Moreover, classification models derived from laser-scanned data often lack semantic depth, requiring significant manual effort to convert surface geometry into structured BIM elements [35,36]. Automated classification systems attempt to bridge this gap, yet integrating contextual awareness into classification models remains an ongoing research challenge. BIM object classification plays a foundational role in semantic enrichment by enabling enhanced data interoperability, improving design analysis, and facilitating better integration with domain-specific applications.

### 2.3. Rule-based and ML-based approaches

The integration of rule-based methods and ML techniques has significantly advanced semantic enrichment in BIM, addressing challenges in automation, classification, and compliance verification. Rule-based methods rely on predefined logic and rules for data analysis and inference, with applications categorized into plug-in tools, object-oriented systems, logical frameworks, and ontological methodologies [37]. Among them, plug-in applications like the Solibri Model Checker [38] analyze design deficiencies and ensure compliance with hard-coded rules. However, the rigidity of such systems, coupled with the complexity of adding new rules, limits their adaptability to diverse scenarios [39].

Object-oriented approaches, such as CORENET e-PlanCheck, extend data extraction and rule processing, allowing for structured compliance checking [3]. Logical frameworks, including decision tables and graphical reasoning models, further refine the rule-based analysis by systematically mapping criteria to compliance decisions [17,40]. Researchers have employed rule-inferencing techniques using predefined ‘IF-THEN’ statements to classify room types [13]. Similarly, rule-based computations have been applied to identify security wall footprint overlaps in multi-story buildings [5]. However, these approaches often require extensive domain knowledge and manual configuration, limiting scalability.

Ontological frameworks have introduced semantic web technologies to overcome some limitations of conventional rule-based systems. By leveraging resource description frameworks (RDF) and web ontology languages (OWL), these approaches enhance data integration and reasoning capabilities, allowing more dynamic compliance verification [41]. Semi-automated ontology generation has also been explored using IfcOWL to improve efficiency and accuracy in regulatory compliance checking, though further refinements are required to address ambiguous or unforeseen scenarios [42].

ML-based approaches present an alternative paradigm, enabling automated learning from data patterns and extending the capabilities of rule-based reasoning. Early ML applications in BIM focused on classifying structural elements and spaces based on geometric and spatial attributes [30]. Room classification remains a critical ML task, as rule-based systems often struggle with variability in spatial definitions. ANNs and graph-based models, including GCNs, have demonstrated effectiveness in capturing interrelationships between walls, doors, and virtual boundaries, improving classification accuracy [13,43].

Graph neural networks (GNNs) further expand the scope of ML in BIM by structuring models as hierarchical graphs, enabling efficient aggregation of spatial attributes from neighboring nodes for classification and reasoning tasks [10]. These applications extend across the BIM lifecycle, encompassing design, construction, and operations. From classifying building shapes during planning to assessing construction quality and predicting energy consumption during operations, GNNs demonstrate versatility and efficiency [44]. GNNs have been utilized for automatic building envelope segmentation, point-cloud data analysis, and maintaining consistency in BIM models [45]. GNNs have also been applied for design reviews, building element classification, and BIM version control, demonstrating their role in addressing interoperability challenges [11,46,47]. These methods represent a significant advancement over traditional approaches, addressing interoperability challenges and enabling semantic transfer technologies [48]. However, while ML-based approaches offer significant advancements, challenges such as error liabilities, computational demands, and integration complexities remain.

### 2.4. Gaps in knowledge

Despite advancements in semantic enrichment, existing methods for BIM-based compliance verification remain limited in addressing complex regulatory requirements. Rule-based methods rely on predefined conditions but struggle with contextual dependencies, while ML models excel at recognizing spatial patterns but require manual validation to mitigate misclassification risks. A hybrid approach that amalgamates the strengths of both methods offers a promising pathway. Prior studies have applied hybrid approaches for object classification tasks, primarily using ensemble learning with stacking, combining multiple models in parallel to improve classification accuracy [32,49]. However, the requirement for tenement identification for compliance checking is a sequential hybrid approach, where ML is used in preprocessing for space classification, while rule-based inferencing is applied at the verification stage.

Fig. 1 illustrates the spectrum of hybrid approaches for semantic enrichment in the ACCC process. The grey, blue, and orange colors signify manual, rule-based, and ML-based approaches at a stage. The circular representation signifies sets and subsets. For example, in the right side approach of the spectrum, a part of the ACCC process can be completed by a rule-based system, whereas ML-based approaches can perform the remaining parts. The vertical lines separate the different adaptation types of hybrid approaches in the ACCC process. The left section denotes the current state of application in the industry. The middle and the right sections focus on the adaptation of hybrid approaches, focusing on the preprocessing and verification stages, respectively. The influence of ML-based methods increases from left to right, enhancing adaptability and applicability across varied scenarios. However, this inductive nature can decrease accuracy, raising concerns about liability in cases of false positives or false negatives. On the contrary, fully rule-based, deductive methods are more suitable for review processes within ULBs due to their reliability and predictability.

The proposed hybrid framework in this paper addresses the gaps in the knowledge through the following ways: (1) this study employs a structured hybrid workflow where ML is used in preprocessing and rule-based logic at verification, ensuring sequential enrichment; (2) the proposed method introduces, user-verification for ML applications in the loop to reduce issues regarding liability during verification stage; (3) an extensive feature engineering was performed to extract and create 19 features from the preconstruction stage IFC models with limited information availability; and (4) highlighting the better performance of feature-rich classical ML models, in comparison to computationally expensive neural networks.

## 3. Research methodology

This study investigates the integration of ML and rule-based inferencing to achieve automated semantic enrichment of BIM models. The focus lies on the preconstruction permitting stage, where the availability of detailed information is often constrained. The research addresses the tenement identification problem in high-rise residential buildings as a case study. A tenement refers to a self-contained housing unit equipped with essential amenities such as sanitation, cooking, and living facilities according to regulation 2(IV)(40) of the Development Control and Promotion Regulations 2034 (DCPR) for Greater Mumbai, India. While humans can recognize tenement units by observing spatial layouts and access routes, computational methods require structured patterns for identification. This process is critical for calculating carpet areas, water tank capacities, and parking space requirements. A couple of examples of code requirements to be verified post-enrichment are elaborated in the following paragraph.

'Carpet area' means the net usable floor area of an apartment, excluding the area covered by the external walls, areas under services shafts, exclusive balcony or verandah areas, and exclusive open terrace areas, but includes the area covered by the internal partition walls of the apartment, according to the Real Estate (Regulation and Development) Act, 2016. This value can be calculated from the footprint of the spaces in a tenement. On the other hand, the parking requirement of an apartment-type residential building depends on the number of tenements and the carpet area. For example, one parking space must be allotted for every four tenements with a carpet area of up to 45 sqm, according to Table 21 of Regulation 44 of DCPR 2034. At the same time, one parking space is required for every half tenement with a carpet area exceeding 90 sqm. Thus, the identification of the tenement becomes an essential step in calculating various code requirements at the verification stage. The study utilizes data from 18 residential buildings encompassing 124 tenement units.

### 3.1. Adopting a hybrid approach

According to researchers, the semantic enrichment step can be divided into four different task types [8]. These tasks include extracting implicit information through calculation or classification steps. On the other hand, new conceptual items that were previously not present in the IFC exchange schema can also be generated. These tasks include association and creation steps. In this study, the task of aggregating IFC spaces into tenement units is categorized as an association task. This process employs rule-based methods to convert implicit information into explicit information for compliance verification. Fig. 2 illustrates the multi-layered approach adopted for semantic enrichment, demonstrating how unavailable or incomplete information is systematically enriched until compliance checks can be conducted. The semantic enrichment techniques adopted in this study are the four task types as defined by Bloch and Sacks [8]. In this study, these steps are adopted through an iterative approach for preprocessing and verification stages until the desired level of enrichment is achieved. This approach was specifically applied to the task of tenement identification, where spaces

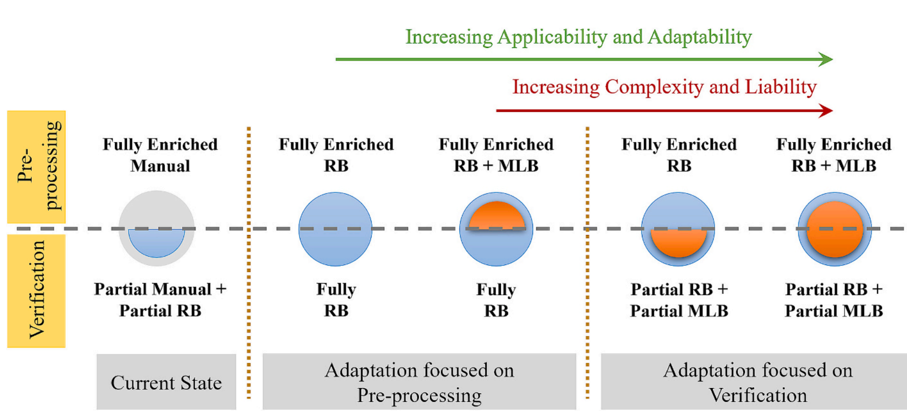


**Fig. 1.** Spectrum of hybrid approaches for semantic enrichment in ACCC.

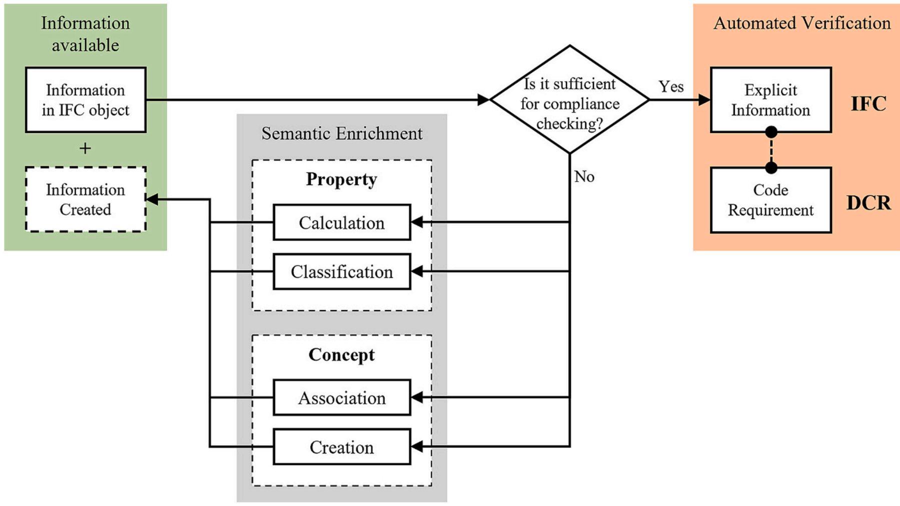


**Fig. 2.** Semantic enrichment task positioning in conversion of implicit information to explicit information.

must first be classified based on their function before being aggregated into cohesive tenement units.

The space identification tasks fall under the broader category of occupancy classification, which is critical for ensuring an accurate representation of spatial elements in BIM models. A typical floor in a residential project may include multiple tenements, each comprising approximately ten distinct rooms, such as a living room, dining room, kitchen, balcony, two bathrooms, two bedrooms, utility room, and passage. Additionally, common areas on each floor often consist of spaces like staircases, corridors, lifts, and ducts. For a 12-story building, this results in over 550 rooms that require precise tagging and categorization. If performed manually, this process becomes even more labor-intensive and error-prone in projects involving multiple buildings. An ML-based approach was adopted to automate space classification tasks to address these challenges.

The hybrid process for space classification and tenement identification task is illustrated in Fig. 3. The workflow begins with extracting a BIM model lacking space labels from the BIM authoring tool. Predicted space labels are generated through feature extraction and classification tasks. Following this, the architect verifies and corrects the predicted labels. Any modifications performed by the user are stored in a database for iterative model improvement. Once the model has been labeled accurately, predefined rule sets are applied to execute the tenement identification process. The enriched model, now containing explicit information, is subsequently utilized for compliance checks. This process map highlights the hybrid approach adopted in this study, wherein semantic enrichment tasks are performed iteratively across the stages of compliance verification. The integration of ML and rule-based techniques ensures a structured, accurate, and automated workflow for tenement identification, streamlining the compliance checking process.

### 3.2. Space classification

The space classification task is a multi-class problem involving more than 20 distinct space types. Hence, it becomes crucial to identify the spaces first, depending on the building type. Mumbai, a metropolitan city in India, has its own development regulations (DCPR), which vary for different types of buildings, such as residential, commercial, industrial etc. Further, mixing different types of buildings increases the associated space types and the total number of spaces to be classified from a single BIM model, thus requiring a bigger pool of training data. However, depending on the availability of training data at the local ULB, this study focused exclusively on multi-story residential buildings in Mumbai to ensure consistency and minimize variability. The dataset was derived from IFC4 schema models, which were extracted from BIM

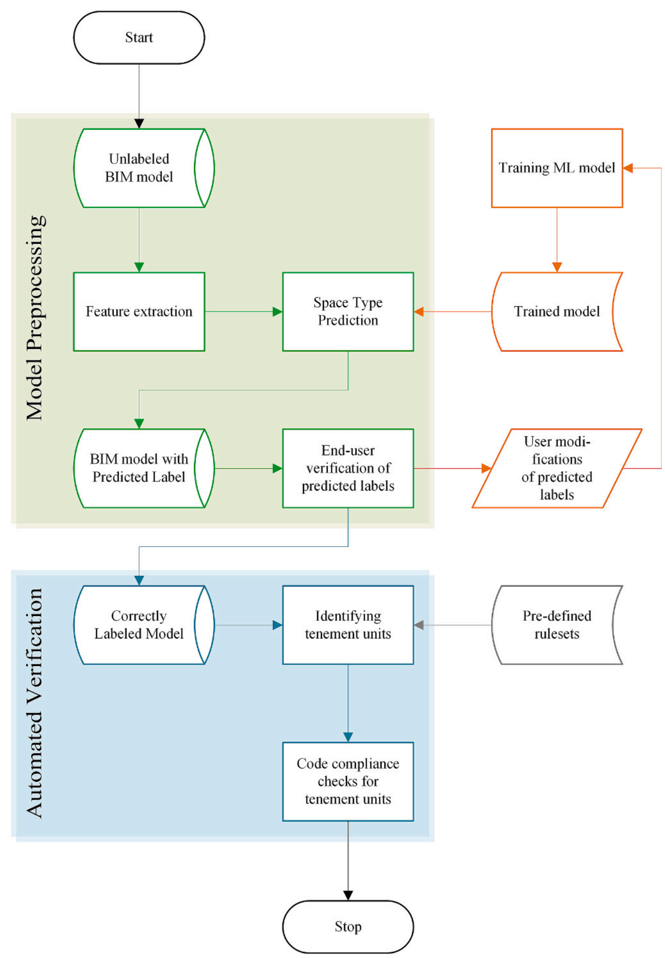


**Fig. 3.** Process map of tenement unit identification for automated code compliance checking adopting a hybrid approach.

authoring tools and represented real-world projects submitted by architects to the local ULB for approval.

#### 3.2.1. Dataset creation

To eliminate redundancy, typical floor plans were excluded from the 18 project models analyzed, as shown in Fig. 4. For classical ML models,

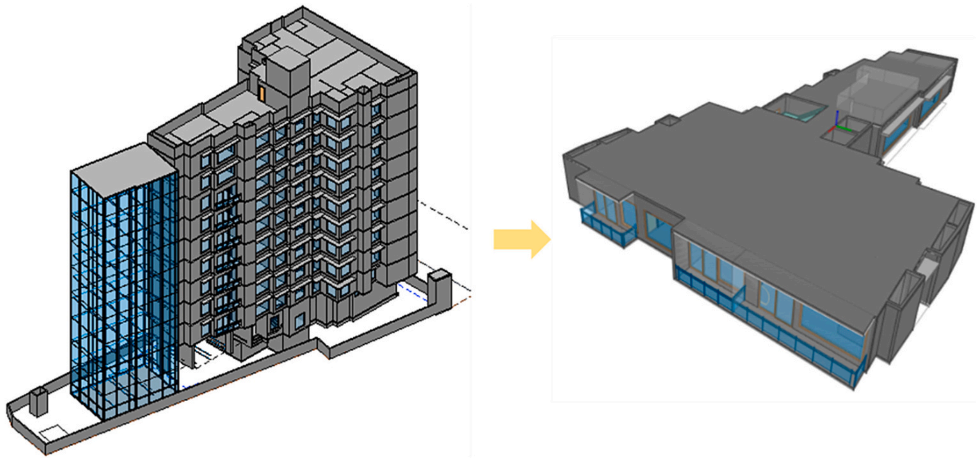

**Fig. 4.** Generating Datapoints from IFC models of multi-story residential buildings.

every room is treated as a data point, whereas for the GNN model, floorplates work as a single data point, with individual spaces represented as nodes and their connections defined as edges. These connections included physical linkages, such as doors, and conceptual relationships, such as open boundaries between spaces. An example of an open connection can be observed between the kitchen, dining, and living rooms, as depicted in Fig. 5, which is a typical floor plan of Indian residential housing with three tenement units.

Table 1 summarizes the space types identified in the project, highlighting an imbalance in data availability across room categories. Very few numbers of datapoints may lead to noise in the training dataset. Therefore, space types with fewer than 30 data points were excluded from the analysis. The final dataset comprised 1269 spaces for classical ML models and 18 graphs for the GNN model. Each graph represented a typical floor plan from the individual projects. The total number of nodes across 18 graphs remains the same as 1269 spaces. However, the graph may vary in size for each building. For example, the building floor plan shown in Fig. 5 consists of 31 nodes and 30 edges representing the connection between these spaces. The connection type can be through doors or open space boundaries, as represented through different colors. Similar to the classical ML models, space types with fewer than 30 data points were not used for GNN model training. Nodes representing these spaces and their connection are represented through dotted lines.

Data imputation was not performed on this dataset to keep the

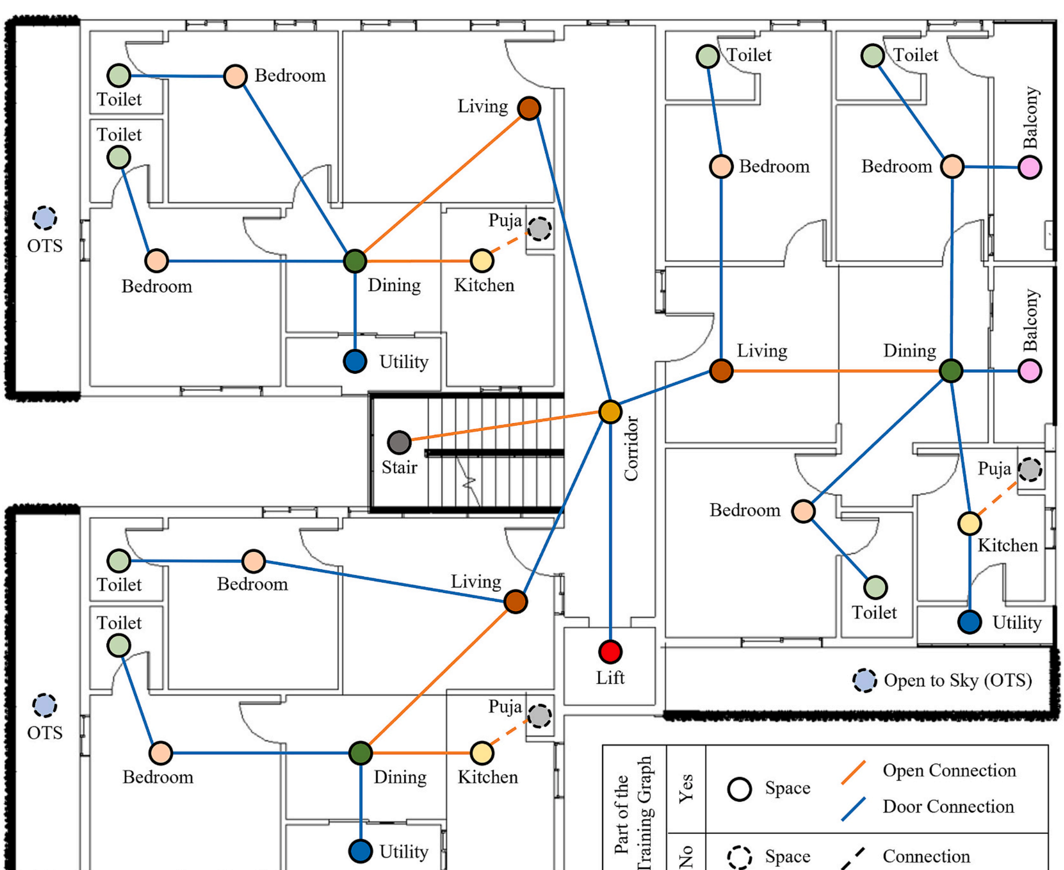


**Fig. 5.** Standard floor plan of a typical Indian residential apartment building.

**Table 1**
Space types, along with their function and sample count.

| Space Type | Description | Count | Considered for Classification |
|---|---|---|---|
| Toilet | A space containing plumbing fixtures along with a water closet and bathing equipment | 283 | Yes |
| Bedroom | A space for resting and sleeping | 242 | Yes |
| Duct | A void space for passing plumbing, electrical, or mechanical services | 132 | Yes |
| Living | A space for social gatherings and activities | 122 | Yes |
| Kitchen | A place for cooking | 120 | Yes |
| Passage | A shared space connecting other spaces of an apartment with the living room | 89 | Yes |
| Lift | A void space used for the movement of lifting equipment | 72 | Yes |
| Balcony | An open space in a tenement enclosed with either short walls or railings and fully closed on at least one side | 56 | Yes |
| Dining | A space used for the consumption of food | 41 | Yes |
| Corridor | A shared space between tenements in a building having access to common staircases and lifts | 40 | Yes |
| Utility | A space used for washing clothes or utensils | 38 | Yes |
| Stair | A space used for vertical movement on feet | 34 | Yes |
| Puja | A space for worship/prayer in a typical Indian household | 15 | No |
| Open to Sky | Open terrace space, but not accessible to the general public | 15 | No |
| Study room | A space used for studying | 9 | No |
| Store room | A space generally attached to a bedroom for clothes storage | 4 | No |
| Entry Hall | A larger entrance space to the apartment (Found in one building with apartments of high carpet area) | 3 | No |
| TV room | A space dedicated explicitly to television (Found in one building with apartments of high carpet area) | 3 | No |

number of data points the same for the classical ML and GNN models. Though the classical models may benefit from addressing the class imbalances through data imputation, the GNN model may loose its inherent topology of the natural relationships by the generation of synthetic edge connections. The imputation is especially tricky as a few of the under-presented nodes in the graph have a high degree of centrality. For example, each project will have a limited number of dining rooms and corridors in the floor plan. Hence, synthetic generation of these central nodes can lead to overfitting. For example, a synthetic corridor node might end up mimicking the same connectivity of the lone present corridor node, as these imputations are performed for each graph individually [50]. Hence, the authors have preferred to keep the unique blueprint of the floor plans of the training graphs for this study.

### *3.2.2. Feature engineering*

The feature vectors for this study were extracted and derived from IFC4 schema models. These feature vectors included eleven input vectors derived directly from IFC object properties and nine additional features derived through calculations. The extracted features are primarily related to space geometry and associated elements, including doors, windows, walls, slabs, and stairs. However, furniture and plumbing fixtures were not considered, as these elements are typically absent in the submitted BIM models during the preconstruction permitting stage.

These feature vectors were identified through interviews conducted with three architects and industry experts. This collaborative process led to the formulation of a comprehensive list of twenty features, summarized in Table 2. These features represent a union of space-identifying criteria proposed by the experts. However, not all features were used in every ML model. In Table 2, the fifth column denotes their application, where ‘C’ indicates usage in classical ML models, ‘GNa’ refers to graph node attributes, and ‘GEa’ applies to graph edge attributes. The third column distinguishes features derived through calculations from those extracted directly from IFC properties, while the fourth column specifies data types, such as continuous, integer, Boolean, or categorical.

For instance, features such as ‘room area’ were directly extracted from IFC properties, whereas attributes like ‘room length’ and ‘room width’ were derived. The ‘length-to-width ratio’ was a particularly critical derived feature, as it helped distinguish common spaces such as corridors from residential rooms. ‘Room length’ was calculated as the maximum distance between parallel boundaries, but in cases where such boundaries were absent, the largest available edge length was used. For circular or ellipsoidal spaces, the diameter or major axis was employed as the length property, with ‘room ‘width’ subsequently derived by dividing the room area by the calculated length.

The number of doors and windows was extracted from second-level IFC relationships. The count is zero if a room has no door or window. Among all connected doors or windows, the largest in size was prioritized as a defining factor. For instance, living rooms typically feature larger entry doors and windows, whereas toilet spaces often have smaller windows with higher sill heights. However, the sill height can not be considered zero in such a case, as zero sill height is possible for large windows. The ‘sill ‘height’ feature was categorized into intervals of 0.5 m to address scenarios where windows were absent, with categories ranging from ‘does not ‘exist’ ‘to’ $>$ 2.5 m.” The minimum sill height among all windows in a space was considered the most critical factor for classification.

For the graph-based model, features such as door height and width were treated as edge attributes, representing connections between two spaces. Additionally, derived properties such as door swing direction were encoded as integer values, with total counts for inward and outward swings calculated. For example, living rooms and corridors often have more outward-swinging doors. Door swing properties were determined from the footprints of door swings in IFC4 models. Double swing doors were counted as both inward and outward swings, while sliding doors were counted as zero in both directions. The door and window materials were not considered features as this information is found to be lacking according to the Level of Development (LOD) of the models.

The LOD framework in BIM is defined as a systematic progression of graphic and semantic information representation [51]. At LOD 100, elements are depicted by generic symbols or massing models that provide only an approximate indication of their existence. This is refined in LOD 200, where elements are represented as generic systems with approximate sizes and locations. At LOD 300, the model elements are sufficiently developed to convey the design intent with accurate size, shape, location, and orientation [24]. At LOD 350, additional information is incorporated to capture the interfaces and connections with adjacent elements, thereby supporting detailed coordination. LOD 400 models are developed to a fabrication-ready stage, complete with detailed assembly and installation data, while LOD 500 represents the as-built condition, offering a field-verified depiction of the constructed elements with exact dimensions and spatial relationships. However, LOD concepts do not relate to the phases of the project’s progress. For example, at a schematic design phase, different model elements can be at different LOD levels [52]. The models submitted for preconstruction permit in this project contained basic floor plans at a schematic level. Hence, generic door and window elements without any material specification are present in the models at the LOD200 level, and the model further lacks details of plumbing and furniture.

On the other hand, for graph networks, spaces can be connected through imaginary separating lines. For example, in Fig. 5, the living area has an open connection with the dining space. A new feature,

**Table 2**
Description of feature vectors and their properties.

| Feature Name | Description | Type of Feature | Type of Data | Used in Models |
|---|---|---|---|---|
| Room area | The footprint area of the space | Property | Continuous | C*, GNa* |
| Room length | Maximum available length in the space | Derived | Continuous | C, GNa |
| Room width | The value derived from dividing room area by room length | Derived | Continuous | C, GNa |
| Room length-to-width ratio | The ratio among space length and width parameters | Derived | Continuous | C, GNa |
| Room height | Clear height above the space below the ceiling | Property | Continuous | C, GNa |
| Number of doors | Total number of doors connected to a room or between two spaces | Property | Integer | C, GNa |
| Door height | The maximum height of all doors associated to a room for classical models.<br>The maximum height among all doors among two spaces for graph models | Property | Continuous | C, GEa* |
| Door width | The maximum width of all doors associated to a room for classical models.<br>The maximum width among all doors among two spaces for graph models | Property | Continuous | C, GEa |
| Door swing in | Total number of doors swinging inwards the space | Derived | Integer | C, GNa |
| Door swing out | Total number of doors swinging outwards from the space | Derived | Integer | C, GNa |
| Number of windows | Total number of windows in a space | Property | Integer | C, GNa |
| Window height | The maximum height among all windows associated with a room | Property | Continuous | C, GNa |
| Window width | The maximum width of all windows associated with a room | Property | Continuous | C, GNa |
| Window sill height | Minimum bottom elevation among windows connected to the space | Property | Categorical | C, GNa |
| Space upper bound | Whether a space is enclosed by a slab or roof above | Derived | Boolean | C, GNa |
| Space lower bound | Whether a space is enclosed by a slab below | Derived | Boolean | C, GNa |
| Walls attached | Total number of walls attached to a space | Property | Integer | C, GNa |
| Short Walls | Count of walls connected to a space that is shorter than 1.5 m | Property | Integer | C, GNa |
| Contains stair | Whether a space overlaps with stair footprints in a floor | Derived | Boolean | C, GNa |
| Spatial connection type | Connection type between two spaces. It can be 'none', 'door', or 'open' | Derived | Categorical | GEa |

[*] Footnote: C = Classical, GNa = Graph Node Attributes, GEa = Graph Edge Attributes.

'spatial connection type', was introduced to capture the spatial relationships in room graph edges. The connection categories considered were 'none', 'door', or 'open'. Boolean features such as 'space upper bound' and 'space lower bound' were also included to indicate whether spaces were enclosed by slabs or roofs, helping to identify void spaces like ducts and lift wells. The IFCOpeningElements associated with IFCSlab are matched with space boundaries to create the Boolean type data for the 'space bound' features. Additionally, short walls, defined as walls shorter than 1.5 m, were identified as an important feature to distinguish the standard walls from balcony walls, parapet walls, and walls in open kitchens. The derived feature 'contains stair' was created by overlaying stair footprints within a floor to identify spaces that overlapped with staircases. These features were extracted from models individually and then programmatically merged to create the final input vector for use in ML models.

### 3.2.3. Data preprocessing for ML algorithms

Prior to applying machine learning models, the dataset was refined by removing space classes with fewer than 30 instances. This step was essential to eliminate noise and ensure the reliability of the classification process. The remaining spaces were then label-encoded to prepare them for ML processing. The final classification dataset comprised 12 distinct space classes. The dataset was randomly divided into three subsets to facilitate model training and evaluation: 70 % for training, 20 % for testing, and 10 % for validation.

The split was performed for the room graph dataset based on project samples, with 12, 4, and 2 projects assigned to the training, testing, and validation sets, respectively. The same project distribution was maintained when creating feature vectors for classical ML models, ensuring consistency across methodologies and allowing for a direct comparison of results. The selected test projects were verified to include all 12 space classes, ensuring a comprehensive evaluation of the classification models. Additionally, the categorical feature 'sill height' was one-hot encoded.

To address potential collinearity among numerical features, correlations between 18 numerical attributes were analyzed using the Spearman correlation method, as depicted in Fig. 6. Features with correlation coefficients exceeding 0.8 were identified, and their impact on model performance was assessed by testing the best-performing model with combinations of these features removed. However, the high correlation between 'window 'height' and 'window 'width' was retained. This decision was based on the observation that this relationship was

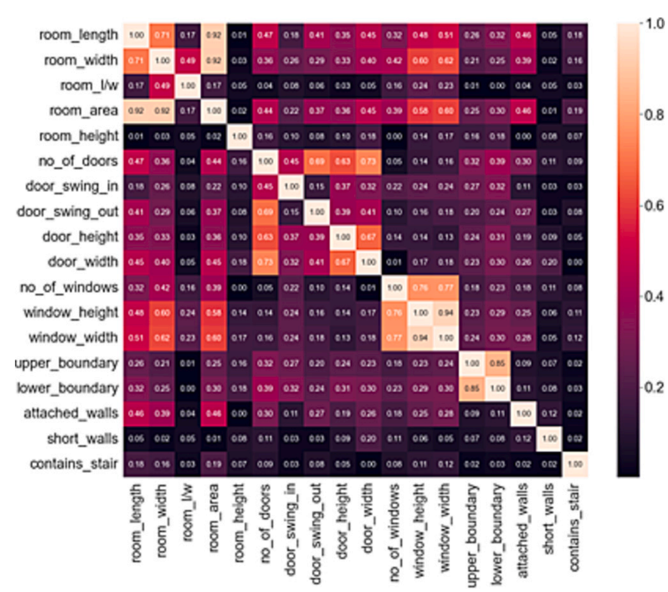


**Fig. 6.** Spearman correlation between numerical features.

likely specific to the dataset and reflective of the architectural design choices; however, in other designs, the window dimensions can vary depending on the project requirements.

#### 3.2.4. Selection of ML models

This study utilized seven supervised classifier models, categorized into state-of-the-art classical models and deep neural networks. The classical models implemented included (i) Support Vector Machines (SVM), (ii) K-Nearest Neighbors (KNN), (iii) Random Forest (RF), (iv) Extreme Gradient Boosting (XGB), and (v) Voting Ensemble (VE).

The SVM model was trained to find optimal hyperplanes in N-dimensional space, maximizing the separation between classes. Hyperparameters, including kernel type, regularization parameter (C), and polynomial degree, were fine-tuned to achieve the best classification performance. The non-parametric KNN classifier relied on proximity measures between data points for prediction, with the number of neighbors specified as its primary hyperparameter. RF, a robust ensemble learning method, was constructed using multiple decision trees, with hyperparameters such as maximum tree depth and the number of estimators optimized during training. Boosting, represented by the XGB model, leveraged a sequence of weak learners, primarily decision trees, to focus on incorrectly predicted samples iteratively. Hyperparameters for XGB, including gamma, maximum depth, and the number of estimators, were carefully tuned to enhance model performance. Finally, the VE algorithm combined the predictions of the top-performing classifiers using a soft voting mechanism based on the probability of outputs from individual models.

Deep neural networks imitate the structure of the human brain, where the information is processed in multiple hidden layers between the input and output layers. To produce non-linearly, activation functions are used between the layers. This study's artificial neural network (ANN) architecture had four hidden layers with 'Sigmoid' activation. The 'SoftMax' activation function was employed before the output layer to handle the multi-class classification task. Regularization was introduced through a dropout layer with a 0.25 probability, preventing overfitting. The batch size used for the sample filtration is 10, and the epoch size is 300 with Adam Optimizer and a Learning rate of 0.0005. The loss function used for training was 'categorical cross-entropy', and the input features were the same as those utilized for classical models.

For the room graph dataset, a modified version of the GraphSAGE algorithm, known as SAGE-E, was adopted, following the methodology proposed by [10]. Unlike conventional GraphSAGE, which employs edges only for message passing during training, SAGE-E incorporated edge properties, iteratively aggregating spatial relationships between connected nodes. The model was trained using the Adam optimizer with a learning rate of 0.005 and a batch size of 1. The training process spanned 200 epochs, with cross-entropy loss employed as the loss function. The SAGE-E model was trained over 'ReLU' activation functions and a dropout probability of 0.2 for regularization.

#### 3.2.5. Metrics of performance evaluation

The models employed in this study were evaluated using the F1-score and accuracy metrics. The F1-score, a widely utilized performance measure, is computed as the harmonic mean of precision and recall scores [53]. This metric is particularly effective for imbalanced datasets, as it accounts for both false positives and false negatives. For the multi-class space classification problem addressed in this study, a weighted F1-score was adopted, as both training and testing models have similar space distributions. This approach ensured that each class contributed to the overall score proportionally to its presence in the dataset. In addition to the F1-score, accuracy was also used as a performance metric, defined as the percentage of correctly classified test samples. While class imbalances can impact accuracy, the F1-score captures the performance of the minority classes. However, accuracy was included as a supplementary metric to compare general correctness between models. These metrics collectively ensured a comprehensive evaluation of the classification models.

### 3.3. Tenement identification

After completing the space type classification, the system requires the user to verify the results, as ML models generate probabilistic outputs that are not entirely accurate. Consequently, the space classification task is designed to serve as a design support system, where user verification and corrections ensure the reliability of subsequent processes. Once verified, the model proceeds to the tenement identification task. Unlike space classification, this semantic task employs a rule-based approach and does not require further user intervention, making it usable in both preprocessing and automated verification stages.

The tenement identification process begins by segregating spaces into 'tenement space' and 'non-tenement space' categories based on their labels. Non-tenement spaces typically include corridors, staircases, lifts, ducts, and balconies. The rule-based methodology focuses on identifying access routes and connectivity among tenement spaces within an apartment. The algorithm first identifies spaces that can be part of a tenement and only filters these spaces for further processing. Next, the space boundaries and associated doors to the filtered spaces are calculated. The connected space pairs are determined by examining shared doors or open boundaries between two rooms, as demonstrated in Algorithm 1. Then, a graph is generated based on the space pairs. As the non-tenement spaces are not part of the created graph, a separate cluster of space units can be extracted. For example, if we drop the 'corridor', 'lift', 'stair', and 'open to sky' nodes from Fig. 5, the graph will only remain with three separate clusters of nodes. The next rule checks the containment of at least one bedroom, one kitchen, and one bathroom in each cluster. If this check succeeds, the spaces cluster is marked and stored as a tenement. For the final step, unique scenarios such as a duplex are checked, where a stair can be part of a dwelling unit, making it a multi-story tenement case.

An associative rule-based approach was adopted for the tenement identification task as the identification method for tenement is deterministic, such as door connectivity or open connectivity. Further, the minimum requirement for tenement is well defined. The tenement configuration or irregular shapes do not impact the algorithm. However, the algorithm depends on the correctly defined list of 'tenement spaces'. Whichever space label is absent from the list is ignored for the tenement identification process, thus creating disconnected clusters. Hence, the essential step for this algorithm is to define the 'tenement spaces'. This list can vary from country to country or region to region, depending on that location's basic dwelling unit configurations. However, once this list is updated according to the local requirements, the algorithm should produce correct outputs across input data types. This method is simpler to implement than ML models, as ML models should be retrained on the local dataset to capture the localized context.

```
input  : IFC model with correctly labeled spaces
output : all_tenements (list of tenements with unique space identifiers)
tenement_spaces ← list of unique space labels associated to a tenement
                  (e.g., ['bedroom', 'toilet', 'living', ... , 'kitchen'])
all_tenements ← empty collection
// Retrieve spaces in the building story and filter based on tenement_spaces
for each story in IFC model stories:
    filtered_spaces ← retrieve spaces in the story and filter by names in
    tenement_spaces
    // Preprocess each space: compute geometry and door
    for each space in filtered_spaces:
        polygon[space] ← generate space polygon footprint from space geometry
        doors[space] ← get list of connected doors for space
    end
    // Generate all possible space pairs
    pairs ← create all unique pairs from filtered_spaces
    connected_pairs ← empty list
    //Check for connections between each pair
    for each (space1, space2) in pairs:
        if polygon[space1] touches polygon[space2] or
          (doors[space1] ∩ doors[space2]) is not empty:
            add (space1, space2) to connected_pairs
        end
    end
    // Group spaces into units using graph connectivity
    graph ← create graph with spaces as nodes and connected_pairs as edges
    units ← identify connected clusters in graph
    // Detect tenement units according to the definition of tenement
    for each unit in units:
        if unit contains at least one bedroom, one kitchen, and one toilet:
            add unit to all_tenements
        end
    end
end
// Detection for special cases like duplex
for tenement in all_tenements:
    polygon[tenement] ← generate tenement polygon footprint from space
    geometry
    for each stair in IFC model stairs on the same story:
        polygon[stair] ← generate stair polygon footprint from space geometry
        if polygon[stair] in polygon[tenement]:
            merge tenement with tenement in the above story with an intersecting
            footprint
        end
    end
    update all_tenements list
end
```

**Algorithm 1.**

## 4. Results

The semantic enrichment task was evaluated using four BIM models from real-life projects, randomly selected during the preparation of the testing dataset. In this section, the performance of various ML models is compared. In addition, the impact of features in models' prediction is observed.

### 4.1. Performance comparison between models

Table 3 presents the comparative performance of various ML models tested on a dataset comprising 255 data points from four apartment buildings.

The voting ensemble model emerged as the best-performing classifier, achieving an F1-score of 0.84 and an accuracy score of 0.86. This ensemble model was constructed using the three highest-performing classical models, SVM, RF, and XGB, each optimized through hyperparameter tuning. The optimal parameters for these algorithms are listed in Table 3. Among individual classifiers, the XGB model demonstrated superior performance, attributed to the ability of hierarchical tree-based models to handle imbalanced datasets effectively. The dataset's features, comprising Boolean, categorical, and integer-type numerical variables with fewer continuous attributes (less than 50 %), contributed to the suboptimal performance of the ANN model.

The SAGE-E algorithm, applied to the room graph dataset, achieved an accuracy score of 0.75 but an F1-score of only 0.65, indicating low recall performance. This limitation can be attributed to the inherent challenges in addressing data imbalance within graph datasets, where resampling interconnected nodes is not feasible. Additionally, the size and structure of room graphs influenced outcomes, with floor-level graphs exhibiting critical connectivity through nodes like corridors, in contrast to less complex tenement-level graphs.

The confusion matrix for the voting ensemble model, shown in Fig. 7, illustrates its performance across all space categories. The x-axis represents the predicted classes, while the y-axis corresponds to the actual classes of the test dataset. The model's poorest performance was observed for utility spaces (67 % accuracy) and dining rooms, which were entirely misclassified. This result suggests that the extracted feature vectors lacked sufficient enrichment to differentiate the dining spaces effectively.

On the other hand, certain space categories, such as ducts and stairs, were consistently classified correctly. Additionally, the model performed reasonably well for balconies, bedrooms, corridors, kitchens, lifts, and toilets, achieving acceptable results for eight out of twelve space categories. The precision for utility spaces, living rooms, and passages could potentially improve with larger training datasets. However, the persistent misclassification of dining rooms across all models underscores the need for further feature engineering.

This misclassification also highlights the challenge of distinguishing between dining rooms, living rooms, and kitchens in 3D models that lack furniture, fixtures, and plumbing equipment (FFPe). Enhancing the feature vectors with FFPe information could significantly improve the ensemble model's performance. This suggests an opportunity for future research into integrating enriched feature sets for better semantic classification in the space classification task.

### 4.2. Feature importance

Alongside algorithms, features play an essential role in determining a

**Table 3**
Performance Comparison of Models on Test Dataset.

| Models | Hyperparameters Tuned | F1-score | Accuracy |
|---|---|---|---|
| SVM | C = 10, degree = 3, kernel Type = 'rbf' | 0.82 | 0.84 |
| KNN | Number of neighbors = 3 | 0.76 | 0.80 |
| RF | Maximum depth = 15, number of estimators = 25 | 0.81 | 0.84 |
| XGB | Gamma = 0.1, maximum depth = 2, number of estimators = 100 | **0.83** | **0.85** |
| VE | Pre-tuned models (SVM + RF + XGB) with the best estimators are used | **0.84** | **0.86** |
| ANN | Number of layers = 4 | 0.47 | 0.60 |
| SAGE-E | Number of layers = 4 | 0.70 | 0.72 |

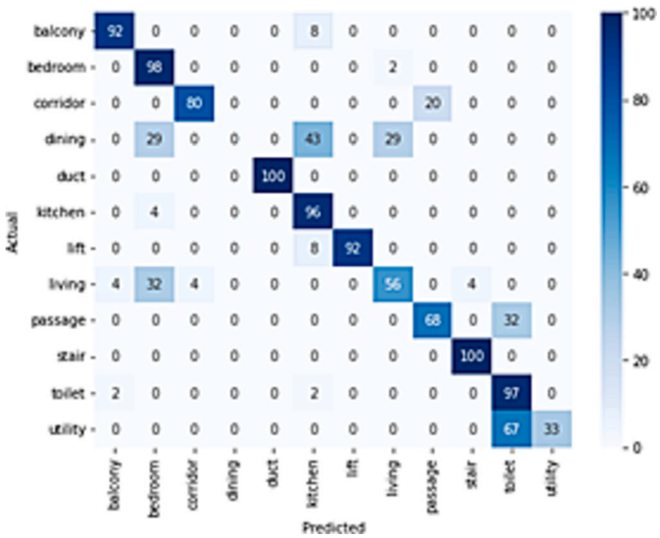


**Fig. 7.** Confusion matrix for Voting Ensemble.

model's performance. The most influential features of the best-performing standalone model, XGB, were analyzed to identify their impact on predictions. Fig. 8 illustrates the top ten features contributing to the performance of the XGB model. Among these, the number of doors swinging inwards and outwards emerged as two of the most significant features for classifying space types. Other key features included the availability of a lower boundary for a space, room area, room width, and the presence of a staircase. High sill heights, specifically in the 1.5 to 2-m range, might have played an important role in classifying toilet spaces. Additional features, such as window width and the number of doors, were also highlighted as essential features for space types. Lastly, the count of short walls might have aided in accurately classifying specific spaces like balconies.

Several iterations were conducted to further explore the contribution of features by removing combinations of features from the feature set and observing the deviation in the XGB model's performance. These iterations, referred to as feature sets (FSn), are summarized in Table 4, where *n* represents the iteration number.

In the first iteration (FS1), features with high correlation values were removed, as shown in Fig. 6. Specifically, 'room area', 'room length', and 'space upper bound' were excluded. This resulted in a decrease in the F1-score and accuracy to 0.81 and 0.82, respectively. In the next iteration (FS2), 'room area' was reinstated to replace 'room width', as it demonstrated higher importance in Fig. 8. This adjustment improved the model's F1-score and accuracy to 0.85 and 0.86, respectively.

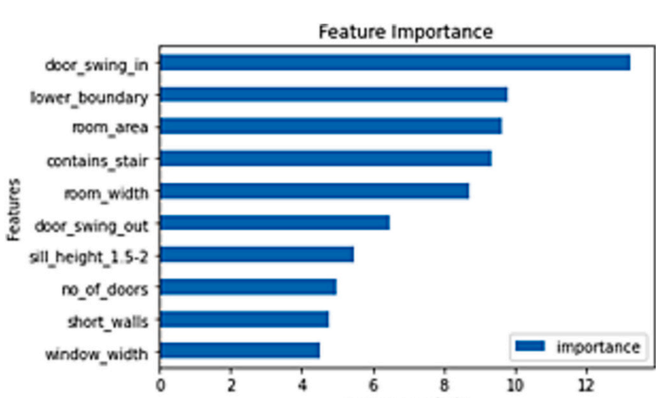


**Fig. 8.** Importance of features in percentage impacting the performance of the XGB model.

**Table 4**
Comparing the effect of features on prediction of space types.

| Iteration | Features dropped | F1-score | Accuracy |
|---|---|---|---|
| FS0 | – | 0.83 | 0.85 |
| FS1 | Room area, room length, space upper bound | 0.81 | 0.82 |
| FS2 | Room length, room width, space upper bound | **0.85** | **0.86** |
| FS3 | Room length, room width | **0.85** | **0.86** |
| FS4 | Other features except the top ten important features identified in iteration FS0 | 0.79 | 0.81 |

In the third iteration (FS3), the 'space upper bound' feature was reintroduced to evaluate its influence on model performance. The results indicated no significant impact on the F1-score or accuracy. Despite this, the inclusion of 'space upper bound' is recommended for larger BIM models, as it may assist in distinguishing ducts from lift rooms, a differentiation that warrants further investigation for practical applications.

The final iteration (FS4) retained only the top ten most important features identified in Fig. 8, removing all other features. This resulted in a notable decline in performance, with the F1-score and accuracy dropping to 0.79 and 0.81, respectively. The results from iterations FS2 and FS3 highlighted that selective pruning of features can enhance model performance. Moreover, when the ensemble model was applied using the feature sets from FS2 and FS3, the F1-score and accuracy further improved to 0.85 and 0.87, respectively.

Apart from selecting features depending on importance values and correlation between features, feature groups were also created depending on the type of features. These feature set combinations, and their performances were evaluated for the XGB and SAGE-E models and are showcased in Appendix A.

### 4.3. Implementation

The outputs from the space classification task were stored in a machine-readable format compatible with BIM authoring tools, allowing for seamless integration and end-user verification. As depicted in Fig. 9, the results were integrated into a building model created using Autodesk Revit for this study. The Dynamo plug-in was employed to update the model with the classification results. After verification and updates by the user, the enriched BIM model was exported in the IFC4 schema format. This enriched model served as the input for the tenement identification task during the verification stage, ensuring the continuity of semantic enrichment across successive processes.

### 4.4. Tenement identification task

Following the room classification task, a 3D BIM model with verified and corrected labels was selected for the tenement identification task. The model represented an apartment floor plate comprising three tenement units interconnected by a corridor. The rule-based algorithm applied during this stage successfully identified all tenement units and associated rooms using the association method detailed in subsection 3.3.

A graphical representation of the tenement identification task is shown in Fig. 10. Further, to check the generalizability of the model, a synthetic model was developed with irregular unit shapes, and the 'kitchen' area was wrongly classified as 'dining' as depicted in Fig. 11. Here the rule-based algorithm successfully marked the irregularly shaped tenement, however, it ignored the regular shaped spaces, as it lacked 'kitchen' unit in the tenement according to its definition. This result demonstrates the successful application of semantic enrichment through a hybrid approach, systematically addressing implicit information at each stage of the process.

The complete pipeline of the proposed hybrid approach has been


**Fig. 9.** Visualization of space type prediction in a BIM authoring tool for verification from end-user.

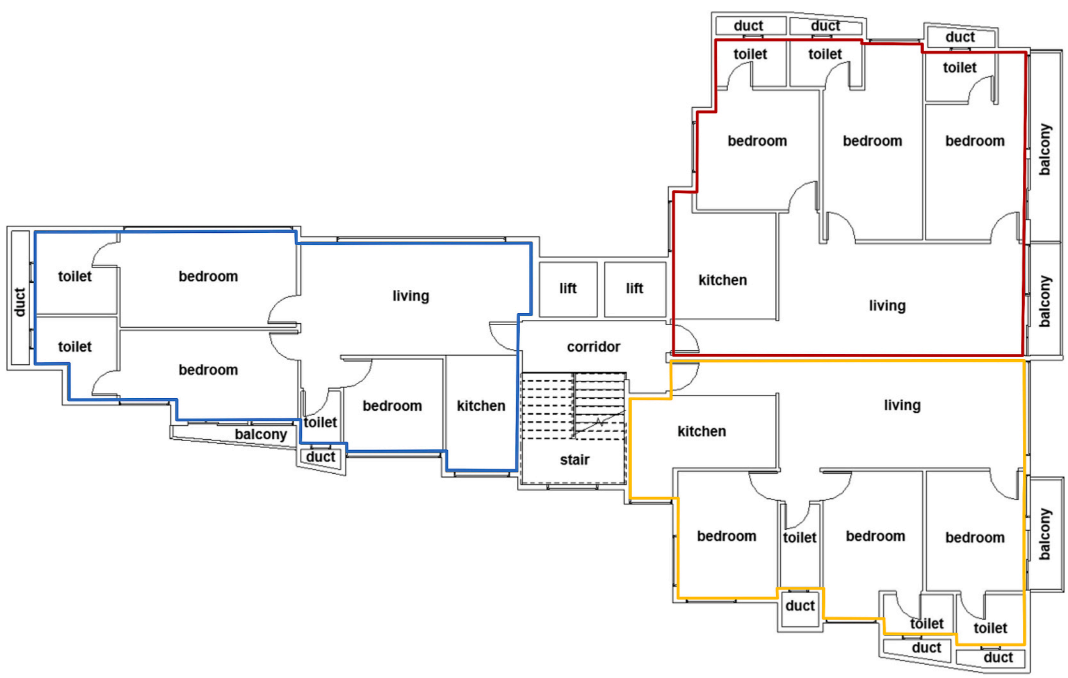


**Fig. 10.** Graphical representation of result from identification of tenement units.

implemented on a pilot basis by an organization working for the local body of the city of Mumbai in India. The pilot implementation is exposed to the end users, usually architects in the city, where we have received positive comments indicating the validity, usefulness, and feasibility of the approach for local urban bodies. Major advantages noted were regarding the time savings as noted by one of the architects in feedback - *"The time taken to complete the task was extremely less compared to when done manually, and this makes the job easier"*. The general feedback was that such interventions would be useful for large-scale projects. The major advantage flagged by users is that even though room classification has produced errors owing to the accuracy inherent in the ML model developed, it was easier and more efficient to correct errors on an ML-generated scheme than tagging all spaces manually, indicating the likeliness of adoption. On the other hand, the tenement identification tasks produced zero errors, thus reducing the chances of human errors in manually marking tenements. Further, the tenement identification approach developed is being used by the ULB to generate carpet areas and parking details to aid automated compliance checking with the city's existing DCPR.

## 5. Discussion

This paper contributes to semantic enrichment in BIM by developing a sequential staged hybrid model, where ML is applied in preprocessing for space classification and rule-based inferencing is used at the verification stage for tenement identification. From a theoretical perspective,

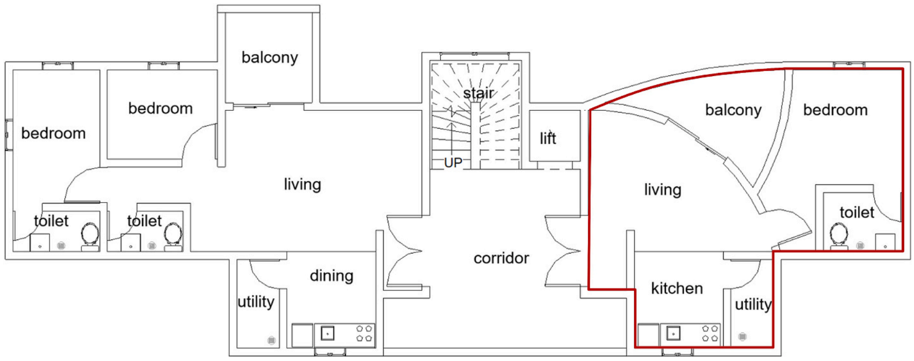


**Fig. 11.** Tenement identification algorithm performance in irregularly shaped and misclassified spaces.

this study demonstrates how a staged hybrid approach can be adopted for complex semantic enrichment problems. Additionally, it highlights the impact of feature engineering, showing how features extracted and derived from IFC models provide better classification performance without relying on computationally expensive deep learning models. From a practical standpoint, the hybrid methodology enhances applicability by bringing the user into the loop and allowing verification after ML preprocessing to reduce liability in compliance verification workflows. The system is designed for scalability and interoperability, making it applicable to information-deficient BIM models. Furthermore, the 19 derived features are comprehensively described and validated, offering a structured reference for future applications in BIM-based semantic enrichment. The methodology has been tested on real-world residential projects in Mumbai, demonstrating its effectiveness and industry relevance.

This study focuses on residential buildings, as the concept of 'tenement units' is intrinsically associated with residential apartments. Consequently, the methodology is not directly applicable to commercial buildings. However, in the case of mixed-use developments, the framework can be effectively adapted by appropriately defining the 'tenement spaces' list. When this list is accurately curated, the ruleset will exclude commercial and public areas, grouping only those spaces designated as tenements, thereby ensuring correct identification.

The framework demonstrates scalability and generalizability across different regional contexts by allowing the modification of the 'tenement spaces' list to align with local regulatory definitions and practices. For instance, in Mumbai, a 'balcony' is typically not considered part of a tenement unit, whereas in other cities, such as those implementing slum rehabilitation schemes, a 'balcony' may be included. The framework remains robust and adaptable by adjusting the 'tenement spaces' list to reflect contextual differences—such as including 'balcony' in the list for specific projects. This flexibility enables the tenement identification ruleset to be reliably applied across diverse geographical regions and regulatory frameworks, thereby ensuring its broader applicability in various urban development and planning scenarios.

This study also found that selective feature pruning improved the F1-score and classification accuracy, underscoring the importance of iterative feature refinement in semantic enrichment tasks. The results demonstrated that a LOD200 BIM model is sufficient to achieve an F1-score of 0.85 through targeted feature engineering, rendering it effective for preconstruction permit stage designs. Nonetheless, higher LOD is recommended for achieving improved predictive accuracy. For instance, incorporating objects such as furniture into the model can reduce misclassifications between living and dining rooms or between living rooms and bedrooms. Similarly, the inclusion of plumbing fixtures in the model may mitigate false predictions between toilets and passages, enhancing classification reliability.

Among the models tested, the ensemble approach emerged as the best performer, effectively handling class imbalances without requiring synthetic data imputation. The hybrid methodology applied at different stages of the model submission process proved instrumental in achieving the desired outcomes. This approach reduces the manual effort required to align models with automated verification requirements by aiding designers during the preprocessing stage. Moreover, user verification of outputs minimizes error propagation and addresses information liability concerns within the system. These advantages contribute to the industry's broader adoption of automated code compliance systems. The hybrid approach demonstrates versatility by leveraging ML for pattern recognition and applying rule-based techniques for extracting implicit information. It offers greater precision than standalone ML modules and is well-suited for handling subjectivity in complex rule applications.

However, the study acknowledges certain limitations. A key challenge in adopting ML lies in the case-specific nature of feature vector generation. Every new application necessitates a unique dataset and feature vector creation, requiring significant time and expertise. Developing extraction logic for derived features often involves close collaboration between domain experts and software developers. To address this, historical data and knowledge repositories across construction design projects could serve as resources for feature development. Additionally, the application of Large Language Models (LLMs) could enhance the efficiency and scalability of feature generation for future studies.

The features identified in this study were derived from narrative inquiries with domain experts and further refined based on the authors' insights from related works. However, some potentially valuable features may have been overlooked, as indicated by the model's performance. The study's features were tailored to align with the Indian architectural design context, which may limit their applicability to projects in different architectural and cultural settings. This highlights the need for region-specific verification of the model in future research.

Another limitation concerns the availability of BIM models. Data insufficiency necessitated the exclusion of space categories with fewer than 30 instances (e.g., study rooms, open-to-sky spaces). Consequently, these spaces will be misclassified consistently if present in the testing model. This limitation extends to any space type infrequently used in building projects. A potential solution involves grouping infrequent spaces into a miscellaneous class to minimize misclassifications. However, the variability in features within this category requires careful consideration to ensure accurate representation. Therefore, while this study demonstrates the efficacy of a hybrid approach to semantic enrichment, its findings also highlight several avenues for future research. First, the proposed approach can easily be adapted and scaled to different scenarios by developing context-specific classification models trained on distinct building typologies, enabling broader applicability across urban contexts. Second, the framework demonstrates its capability to utilize LOD200 models. This can further be enhanced as more and more users submit LOD300 models for scrutiny, and such data would be available for the model to learn further. Such enriched models, when coupled with the use of emerging AI technologies, such as LLMs and automated feature extraction pipelines, can integrate the context more robustly to support intelligent rule generation and reduce expert intervention in feature engineering.

## 6. Conclusion

This paper presented the application of a hybrid approach to the semantic enrichment of BIM models, combining machine learning and rule-based techniques in sequence to address the tenement identification problem. The semantic enrichment process was demonstrated as a stepwise methodology applicable across the design lifecycle. Automating the creation and tagging of model objects significantly reduces the preprocessing efforts required to achieve automated compliance checking. Furthermore, the enriched models facilitate straightforward rule-inferencing, thereby enhancing the success rate of design verification during compliance evaluation. This method decreases rejection rates and redesign cycles, substantially improving overall process efficiency.

The findings from the ML application in the model preprocessing phase indicated that ensemble models effectively handle space-type class imbalances. Additionally, feature engineering was found to play a pivotal role in improving model performance, as derived features exerted a significant influence on predictive accuracy. The study also revealed that ML algorithms, such as the boosting and decision tree models, outperform neural networks for datasets with a higher proportion of discrete variables.

Rule-based associations were employed in the automated verification phase to identify tenement units. These associations proved flawless across all test models, as the foundational space type classification was successfully completed during the preprocessing stage. When ML models are applied, the need for user verification emerges as a critical safeguard. Therefore, this research advocates using ML techniques during preprocessing to mitigate the risk of false positives and false negatives within automated compliance checking systems. The suitability of semantic enrichment methods depends on the nature of the task: ML-based approaches are better suited for subjective and complex classifications, such as spaces, while rule-based inferencing is more effective for clearly defined tasks, such as identifying tenement groups or sunshade projections [14].

This study focused on a single case example to illustrate the potential of hybrid approaches in automated code compliance checking. However, the proposed framework can be extended to other semantic enrichment scenarios to evaluate its broader applicability. The primary aim of this framework is to support end-users, such as designers and architects, by streamlining the model submission process. This approach reduces the manual effort required for preprocessing and enhances system productivity, thereby benefiting urban local bodies by decreasing application processing times. Future research in user-centric semantic enrichment will further reinforce the adoption of automated code compliance checking systems, enabling their seamless integration into industry practices.

## CRediT authorship contribution statement

**Ankan Karmakar:** Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Venkata Santosh Kumar Delhi:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Acknowledgements

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## Appendix A. Appendix

The feature vectors created for the project were segregated into groups based on their types, as shown in Table 5. The performance based on the combination of these groups is presented in Table 6 and Table 7 for the best-performing XGB model and the SAGE-E model proposed by Wang et al. [10]. As feature vectors vary from existing literature, a direct performance comparison between the two ML models cannot be performed. The state-of-the-art SAGE-E model was trained on features extracted from images, where important edge features such as door material were present. However, in this study, the BIM model available lacked such data at the preconstruction permit stage. On the other hand, some additional features, such as door swings and sill height, are not available in the RoomGraph dataset created for the SAGE-E model.

The best-performing XGB model according to feature groups was iteration X0 in Table 6. However, as shown in Table 4, feature sets FS2 and FS3 curated based on feature importance and correlation matrix have shown a better performance. On the other hand, for the SAGE-E model, the model's performance peaked in iteration S5 of Table 7, where only door width and height features were dropped from the edge attributes. The performance of the SAGE-E model was further analyzed by changing the number of layers from 3 to 5. However, the model performed best with 4 SAGE-E layers, as shown in Table 8.

**Table 5**
Details of feature groups.

| Group ID | Feature Group | Feature Name |
|---|---|---|
| FG1 | Space dimensions | Room area |
| | | Room length |
| | | Room width |
| | | Room length-to-width ratio |
| | | Room height |
| FG2 | Space boundaries | Space upper bound |
| | | Space lower bound |
| | | Walls attached |

(*continued on next page*)

**Table 5** (*continued*)

| Group ID | Feature Group | Feature Name |
|---|---|---|
| | | Short walls |
| FG3 | Space connection type | Spatial connection type – opening |
| | | Spatial connection type – door |
| FG4 | Space inclusions | Contains stair |
| FG5 | Door features (set 1) | Number of doors |
| | | Door swing in |
| | | Door swing out |
| FG6 | Door features (set 2) | Door height |
| | | Door width |
| FG7 | Window features | Number of windows |
| | | Window height |
| | | Window width |
| | | Window sill height |

**Table 6**
Comparing XGB model performance based on feature sets.

| Iterations | Features Dropped | F1-score | Accuracy |
|---|---|---|---|
| X0 | – | **0.83** | **0.85** |
| X1 | FG1 | 0.71 | 0.73 |
| X2 | FG2 | 0.67 | 0.70 |
| X3 | FG6 | 0.82 | 0.84 |
| X4 | FG7 | 0.82 | 0.83 |
| X5 | FG5, FG6 | 0.74 | 0.76 |
| X6 | FG5, FG6, FG7 | 0.71 | 0.74 |
| X7 | FG6, FG7 | 0.81 | 0.83 |
| X8 | FG1, FG2 | 0.67 | 0.70 |
| X9 | FG1, FG5, FG6, FG7 | 0.36 | 0.43 |
| X10 | FG2, FG5, FG6, FG7 | 0.67 | 0.69 |

**Table 7**
Comparing SAGE-E model performance based on feature sets.

| Iterations | Features Dropped | F1-score | Accuracy |
|---|---|---|---|
| S0 | – | 0.70 | 0.72 |
| S1 | FG1 | 0.48 | 0.56 |
| S2 | FG2 | 0.65 | 0.73 |
| S3 | FG3 | 0.54 | 0.62 |
| S4 | FG5 | 0.66 | 0.67 |
| S5 | FG6 | **0.73** | **0.75** |
| S6 | FG7 | 0.58 | 0.66 |
| S7 | FG5, FG6 | 0.67 | 0.71 |
| S8 | FG3, FG5, FG6 | 0.48 | 0.55 |
| S9 | FG6, FG7 | 0.65 | 0.69 |
| S10 | FG3, FG5 | 0.66 | 0.70 |
| S11 | FG3, FG6 | 0.65 | 0.65 |
| S12 | FG6, FG1 | 0.52 | 0.53 |
| S13 | FG6, FG2 | 0.71 | 0.72 |
| S14 | FG6, FG1, FG2 | 0.51 | 0.57 |
| S15 | FG6, FG1, FG2, FG7 | 0.50 | 0.57 |

**Table 8**
Performance of SAGE-E variants.

| Layers | F1-score | Accuracy |
|---|---|---|
| 3 | 0.66 | 0.70 |
| 4 | 0.73 | 0.75 |
| 5 | 0.59 | 0.68 |

## Data availability

All data, models, or code generated or used during the study are proprietary in nature and, hence, unavailable to access. It may only be provided with restrictions such as anonymized data on request with approval from the research sponsoring organization.

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